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TELESCOPING DRAIN

Abstract

A telescoping riser assembly and/or telescoping riser system can be used to drain water from an agricultural field without hindering agricultural activities performed on the agricultural field and without damaging any aspect(s) of the assembly. The telescoping drain riser assembly can include underground piping wherein a cylindrical member of the telescoping riser assembly is configured to selectively telescopically extend from and/or retract into the underground piping. The telescoping riser assembly can further comprise a vibration isolator, a snap cap, and a main cap. The vibration isolator can engage and/or be integrally formed with the underground piping and can slidably interact with the cylindrical member to facilitate the telescopic extension and/or retraction. The main cap can be attached to an end of the cylindrical member via the snap cap. The main cap can fit firmly to the ground when the cylindrical member is fully retracted into the underground piping.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to an apparatus, system, and/or a corresponding method of use/manufacture having applications in at least the agricultural and tile drainage industries. More particularly, but not exclusively, the present disclosure relates to a telescoping drain riser for use as part of a tile drainage operation.

BACKGROUND

[0002] The background description provided herein gives context for the present disclosure. Work of the presently named inventors, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art.

[0003] Tile drainage is a form of drainage used in agriculture to remove excess water from agricultural fields. Excess surface water that accumulates on agricultural fields can be drained via pumping, use of ditches, and/or other methods. Tile drainage is an effective way to drain sub-surface water from agricultural fields as well as water that accumulates on agricultural fields above ground. Tile drainage operations can help control erosion of agricultural fields.

[0004] Tile drainage operations often involve inserting underground tubing/piping beneath the surface of the soil of an agricultural field. Such tubing/piping can be used to drain water from agricultural fields to eliminate and/or mitigate erosion of the ground. Existing tile drainage operations include risers that are often tube-shaped. Such risers extend up from the ground wherein the risers are operatively attached to aspect(s) of the tile drainage operation (which can include the underground tubing/piping) that are located underneath the surface of the ground. Such risers often extend to a height 3 to 5 feet above the surface of the ground. Additionally, a typical agricultural field with a tile drainage system may include 30 risers.

[0005] While the risers extending above the surface of the ground may be beneficial and/or necessary for the tile drainage operation to function properly, such risers extending above the ground creates problems at particular times throughout the agricultural process. For instance, problems arise when planting, harvesting, fertilizing, plowing, and/or performing any other agricultural activity that requires an agricultural vehicle and/or agricultural implement to traverse an agricultural field. Farmers controlling agricultural vehicles/implements and/or autonomous agricultural vehicles/implements are forced to navigate and/or steer around the risers in order to perform an agricultural activity such as planting or harvesting so as not to run into and/or over the risers which could result in damage to the riser and/or to the agricultural equipment. Additionally, during harvest when crops are fully developed, it can be difficult and/or impossible to see the risers which leads to farmers often running into the risers with agricultural equipment resulting in loss of time, effort, and money spent repairing and/or replacing said risers or agricultural equipment. Navigating around such risers leads to an increase in the amount of time needed to perform the agricultural activity, an increase in the effort involved in performing the agricultural activity, and can also lead to lower yields as less seed may be planted and/or it may be more difficult to harvest crops properly due to crop rows being curved and/or missing. An increase in time and effort needed to perform agricultural activities results in greater costs for the farmer based on greater fuel consumption to power the agricultural vehicles/implements for a longer period of time, more money to be paid in wages to hired laborers, as well as other reasons. A decrease in yields leads to less revenue for the farmer. Thus, these risers extending out of the ground can affect a farmer financially.

[0006] With the evolution of agricultural implements getting bigger and bigger and also being able to service more and more rows of crops in wider swaths, the problem of navigating around these risers has only been exacerbated. For example, while 6-row or 12-row planters were perhaps more common years ago, now 16-row and 24-row planters are common as well as even larger planters. Larger agricultural equipment is less maneuverable which causes more issues when navigating around the risers.

[0007] One existing way farmers have dealt with this issue of needing to navigate around risers is to remove the above-ground portion of the risers when performing an agricultural activity. However, when the above-ground portion of the risers is removed (i.e., stripped away from the surface of the ground), an open end of the underground tubing/piping is left exposed. Therefore, if the risers are removed during agricultural activities such as planting and/or harvesting, when the agricultural equipment passes over and/or comes in contact with the exposed underground tubing/piping, said agricultural equipment will damage the exposed tubing/piping. This could result in time, effort, and money needing to be spent to repair and/or replace aspects of the tile drainage operation. Additionally, when the above-ground portion of the risers are removed when performing an agricultural activity, debris is able to enter the exposed end of the underground tubing/piping. Debris entering the underground tubing/piping and/or entering other aspects of the tile drainage operation can negatively affect the functionality and efficiency of the water drainage as well as potentially damage aspects of the tile drainage operation. Thus, debris entering the underground tubing/piping and/or other aspects of the tile drainage operation could result in the need to clean aspects of the tile drainage operation as well as repair and/or replace aspects of the tile drainage operation. Such cleaning, repair, and/or replacement would cost a farmer time, effort, and money. Additionally, if the tile drainage operation is not draining water properly, due to damage and/or buildup of debris, the farmer's yields, and thus the farmer's revenue, would be negatively affected. [0008] Thus, there exists a need in the art for a drain riser assembly that does not need to be navigated around when performing an agricultural activity in an agricultural field. There exists an additional need in the art for a drain riser assembly that does not allow damage to aspect(s) of the tile drainage system or allow debris to enter the tile drainage operation when performing an agricultural activity in an agricultural field.

SUMMARY

[0009] The following objects, features, advantages, aspects, and/or embodiments, are not exhaustive and do not limit the overall disclosure. No single embodiment need provide each and every object, feature, or advantage. Any of the objects, features, advantages, aspects, and/or embodiments disclosed herein can be integrated with one another, either in full or in part.

[0010] It is a primary object, feature, and/or advantage of the present disclosure to improve on or overcome the deficiencies in the art.

[0011] It is a further object, feature, and/or advantage of the present disclosure to provide a telescoping drain riser assembly wherein a top riser can be extended from and retracted into a bottom riser and/or underground piping.

[0012] It is a further object, feature, and/or advantage of the present disclosure to allow for proper water drainage from an agricultural field without hindering the ability to perform agricultural activities on the agricultural field. Aspects and/or embodiments of the present disclosure allow for a top riser to be extended above ground when agricultural activities are not being performed on an agricultural field and retracted below ground when agricultural activities are being performed on an agricultural field. Thus, aspects and/or embodiments of the present disclosure allow for a farmer to not have to navigate around an extended riser when performing agricultural activities. This allows farmers to perform agricultural activities in a more fuel-efficient manner, faster, and with less effort. Therefore, the farmer can limit costs related to fuel consumption and/or payment to hired laborers. Additionally, since farmers and/or operators of agricultural equipment do not have to navigate around extended risers in an agricultural field, rows of crops can remain straight and uninterrupted rather than possibly being curved and/or interrupted. This leads to maximization of yields which leads to higher revenue for the farmer. Thus, aspects and/or embodiments of the present disclosure limit costs and maximize revenue for a farmer.

[0013] It is a further object, feature, and/or advantage of the present disclosure to allow for proper water drainage and the ability to perform agricultural activities without damaging any aspect(s) of the water drainage system and without allowing unwanted debris to enter aspect(s) of the water

drainage system. Aspects and/or embodiments of the present disclosure include a main cap on one end of the top riser wherein the main cap is configured to be flush with the ground and/or configured to fit firmly to the ground when the top riser is fully retracted such that the top riser is underground. The main cap is configured to protect aspect(s) of the drainage system from damage and from debris when agricultural activities are being performed. Thus, farmers do not have to navigate around riser locations for fear of damaging the riser, and/or other aspects of the drainage system, or for fear of causing unwanted debris to enter piping and/or other aspects of the drainage system. Rather, a farmer can perform agricultural activities normally as if the risers were not present at all. Again, this allows farmers to perform agricultural activities in a more fuel-efficient manner, faster, and with less effort. Therefore, the farmer can limit costs related to fuel consumption and/or payment of hired laborers. Further, the main cap protects the riser assembly, and/or other aspects of the drainage system, from damage and unwanted debris. Thus, aspects and/or embodiments of the present disclosure allow a farmer to not have to spend time and/or money repairing and/or replacing aspects of the drainage system. Also, farmers do not have to spend time and/or money removing unwanted debris from the riser assemblies and/or other aspects of the drainage system. Accumulation of debris in the riser assemblies and/or other aspects of the drainage system can cause the drainage system to not function properly, which can lead to even more time, money, and effort needed to repair or replace aspects of the drainage system. Additionally, since farmers and/or operators of agricultural equipment do not have to navigate around riser locations, rows of crops can remain straight and uninterrupted rather than being curved and/or interrupted. This leads to maximization of yields which leads to higher revenue for the farmer. Furthermore, damage and buildup of debris can lead to the drainage system not functioning properly, which can hurt crop yields and, thus, hurt the farmer's revenue. Therefore, by protecting the riser assemblies and/or other aspects of the drainage system from damage and from buildup of debris, aspects and/or embodiments of the present disclosure disclosed herein allow a farmer to maximize revenue. Thus, aspects and/or embodiments of the present disclosure limit costs and maximize revenue for a farmer.

[0014] It is a further object, feature, and/or advantage of the present disclosure to provide a vibration isolator that is configured to slidingly engage the top riser of the drain riser assembly.

[0015] It is a further object, feature, and/or advantage of the present disclosure to provide a locking mechanism on the top riser such that the top riser can be locked in place relative to the vibration isolator and/or to the underground piping.

[0016] It is a further object, feature, and/or advantage of the present disclosure to provide a telescoping riser system comprising one or more telescoping riser assemblies wherein a user can interact with a human-machine interface (HMI) to determine and/or monitor the location of each of the one or more telescoping riser assemblies.

[0017] It is a further object, feature, and/or advantage of the present disclosure to eliminate the need for an orifice in the bottom riser that goes into the ground. The orifice in the bottom riser can restrict water flow and is included only to comply with archaic governmental regulations. While the orifice can still be used in combination with the telescoping riser systems described herein, it is more beneficial to either (i) eliminate the orifice in whole or (ii) move the orifice to the top riser.

[0018] The apparatus(es) disclosed herein can be used in a wide variety of applications. For example, the drain riser assembly disclosed herein could be used as part of any tile drainage system. Additionally, the telescoping riser system disclosed herein could be used in a wide variety of environments other than agricultural fields. For example, such a system could be used for water drainage in any setting such as any field, yard, sports and recreation area, construction site, and the like.

[0019] It is preferred the apparatus be safe, cost effective, and durable. For example, some of the purposes/objectives/advantages of the present disclosure, which are noted above, include minimizing costs and maximizing revenue. Additionally, aspects and/or embodiments of the

present disclosure can be constructed from plastic and/or metal, which provides safety, cost effectiveness, and durability. The apparatus(es) and/or system(s) described herein can be adapted to resist thermal transfer, electric conductivity, and/or failure (e.g. cracking, crumbling, shearing, creeping) due to excessive and/or prolonged exposure to tensile, compressive, and/or balanced forces acting on the apparatus(es).

[0020] At least one embodiment disclosed herein comprises a distinct aesthetic appearance. Ornamental aspects included in such an embodiment can help capture a consumer's attention and/or identify a source of origin of a product being sold. Said ornamental aspects will not impede functionality of the present disclosure.

[0021] Methods can be practiced which facilitate use, manufacture, assembly, maintenance, and repair of an apparatus and/or system which accomplishes some or all of the previously stated objectives.

[0022] Apparatus(es) described herein, including the telescoping riser assembly, can be incorporated into systems which accomplish some or all of the previously stated objectives. Additionally, system(s) described herein, including the telescoping riser system, can be incorporated into larger designs which accomplish some or all of the previously stated objectives.

[0023] According to some aspects of the present disclosure, a drain riser assembly for use with water drainage from an agricultural field comprises: a cylindrical member; a vibration isolator configured to slidably interact with the cylindrical member; a snap cap removably attachable to one end of the cylindrical member; and a main cap removably attachable to the snap cap.

[0024] According to at least some aspects of the present disclosure, the cylindrical member comprises opposing guide slots.

[0025] According to at least some aspects of the present disclosure, the vibration isolator is a removable bushing comprising opposing tabs.

[0026] According to at least some aspects of the present disclosure, the opposing guide slots of the cylindrical member are configured to slidably engage the opposing tabs of the bushing.

[0027] According to at least some aspects of the present disclosure, the bushing comprises a body having an outer surface and an inner surface, further wherein the bushing is configured to engage underground piping via the outer surface, and/or flange(s) thereof, and slidably interact with the cylindrical member via the inner surface.

[0028] According to at least some aspects of the present disclosure, when the bushing is engaged with the underground piping, the cylindrical member is capable of sliding relative to the bushing such that the cylindrical member can telescopically extend from and retract into the underground piping, further wherein a body of the main cap fits firmly to the ground when the main cap is attached to the snap cap, the snap cap is attached to the cylindrical member, and the cylindrical member is fully retracted into the underground piping.

[0029] According to at least some aspects of the present disclosure, the snap cap comprises opposing slots wherein the slots of the snap cap are configured to slidably interact with a back side of the guide slots of the cylindrical member.

[0030] According to at least some aspects of the present disclosure, the snap cap further comprises opposing protrusions and the cylindrical member comprises two or more apertures, further wherein each of the opposing protrusions are configured to engage one of the two or more apertures when the snap cap is attached to the cylindrical member.

[0031] According to at least some aspects of the present disclosure, the cylindrical member comprises a locking mechanism that allows the cylindrical member to be locked in place relative to the vibration isolator.

[0032] According to at least some aspects of the present disclosure, the snap cap comprises an overmold and the main cap comprises a connection member, further wherein the overmold and the connection member can matingly engage to attach the snap cap to the main cap.

[0033] According to at least some aspects of the present disclosure, the cylindrical member

comprises first, second, and third sections, further wherein the first and second sections are separated via a first cut line and the second and third sections are separated via a second cut line.

[0034] According to at least some aspects of the present disclosure, the vibration isolator comprises one or more tabs integrally formed with underground piping.

[0035] According to at least some aspects of the present disclosure, the cylindrical member is capable of sliding relative to the one or more tabs integrally formed with the underground piping such that the cylindrical member can telescopically extend from and retract into the underground piping, further wherein a body of the main cap fits firmly to the ground when the main cap is attached to the snap cap, the snap cap is attached to the cylindrical member, and the cylindrical member is fully retracted into the underground piping.

[0036] According to at least some aspects of the present disclosure, a telescoping riser assembly for use with water drainage from an agricultural field comprises: a bottom riser positioned generally underground; a vibration isolator configured to interact with or be integrally formed with the bottom riser; and a top riser configured to slidably interact with the vibration isolator to extend from and retract into the bottom riser in a telescopic manner.

[0037] According to at least some aspects of the present disclosure, the vibration isolator is a removable bushing configured to interact with the bottom riser.

[0038] According to at least some aspects of the present disclosure, the vibration isolator comprises one or more tabs integrally formed with the bottom riser.

[0039] According to at least some aspects of the present disclosure, the bottom riser and the top riser are generally cylindrical.

[0040] According to at least some aspects of the present disclosure, a telescoping riser system for use with water drainage from an agricultural field comprises: at least one drain riser assembly of claim 1, wherein each of the at least one drain riser assemblies is positioned in an agricultural field.

[0041] According to at least some aspects of the present disclosure, the system further comprises a human-machine interface (HMI) operationally connected to each of the at least one drain riser assemblies.

[0042] According to at least some aspects of the present disclosure, the system further comprises a positioning system operationally connected to each of the at least one drain riser assemblies and/or to the HMI to determine and/or monitor the location of each of the at least one drain riser assemblies.

[0043] These and/or other objects, features, advantages, aspects, and/or embodiments will become apparent to those skilled in the art after reviewing the following brief and detailed descriptions of the drawings. Furthermore, the present disclosure encompasses aspects and/or embodiments not expressly disclosed but which can be understood from a reading of the present disclosure, including at least: (a) combinations of disclosed aspects and/or embodiments and/or (b) reasonable modifications not shown or described.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0044] Several embodiments in which the present disclosure can be practiced are illustrated and described in detail, wherein like reference characters represent like components throughout the several views. The drawings are presented for exemplary purposes and may not be to scale unless otherwise indicated.

[0045] As will be understood, any of the aspects of any of the embodiments shown and/or described herein could be combined with one another to form any number of embodiments, whether expressly disclosed or not, which would be understood by one skilled in the art.

[0046] FIG. 1 shows a front elevation view of aspects of an example drain riser assembly, wherein

a snap cap and main cap are fully attached to a cylindrical member.

[0047] FIG. 2 shows a side elevation view of aspects of the drain riser assembly of FIG. 1, wherein the snap cap and/or main cap are not fully attached to the cylindrical member.

[0048] FIG. 3 shows a zoomed-in view of aspects of the drain riser assembly of FIG. 2, wherein aspect(s) of the snap cap are shown interacting with aspect(s) of the cylindrical member.

[0049] FIG. 4 shows a zoomed-in view of aspects of the drain riser assembly of FIG. 2, wherein aspect(s) of the snap cap are shown interacting with aspect(s) of the main cap.

[0050] FIG. 5 shows a side elevation view of a cylindrical member of a drain riser assembly according to some aspects of the disclosure.

[0051] FIG. 6 shows a zoomed-in view of a locking mechanism of the cylindrical member of FIG. 5.

[0052] FIG. 7 shows a top plan view of the cylindrical member of FIG. 5.

[0053] FIG. 8 shows a zoomed-in view of aspects of the cylindrical member of FIG. 7.

[0054] FIG. 9 shows a perspective view of a bushing according to some aspects of the disclosure.

[0055] FIG. 10 shows a top plan view of the bushing of FIG. 9.

[0056] FIG. 11 shows a side elevation view of the bushing of FIG. 9.

[0057] FIG. 12 shows a side elevation view of underground piping according to some aspects of the disclosure.

[0058] FIG. 13A shows a top plan view of the underground piping of FIG. 12.

[0059] FIG. 13B shows a perspective view generally of the top of underground piping according to some aspects of the disclosure.

[0060] FIG. 14 shows a perspective view of a snap cap according to some aspects of the disclosure.

[0061] FIG. 15 shows a top plan view of the snap cap of FIG. 14.

[0062] FIG. 16 shows a side elevation view of the snap cap of FIG. 14.

[0063] FIG. 17 shows a front elevation view of the snap cap of FIG. 14.

[0064] FIG. 18 shows a perspective view of a main cap according to some aspects of the disclosure.

[0065] FIG. 19 shows a top plan view of the main cap of FIG. 18.

[0066] FIG. 20 shows a side elevation view of the main cap of FIG. 18.

[0067] FIG. 21 shows a perspective view of a bushing and underground piping according to some aspects of the disclosure.

[0068] FIG. 22 shows a perspective view of the bushing of FIG. 21 partially inserted into the underground piping of FIG. 21.

[0069] FIG. 23 shows a perspective view of the bushing of FIG. 21 partially inserted into the underground piping of FIG. 21.

[0070] FIG. 24 shows a perspective view of the bushing of FIG. 21 fully inserted into the underground piping of FIG. 21.

[0071] FIG. 25 shows a perspective view of aspects of an example telescoping riser assembly wherein a cylindrical member is removed from a bushing.

[0072] FIG. 26 shows a perspective view of aspects of the telescoping riser assembly of FIG. 25, wherein the cylindrical member is engaged with the bushing and is locked in the fully extended position.

[0073] FIG. 27 shows a perspective view of aspects of the telescoping riser assembly of FIG. 25, wherein the cylindrical member is partially retracted into the bushing and underground piping.

[0074] FIG. 28 shows a perspective view of aspects of the telescoping riser assembly of FIG. 25, wherein the cylindrical member is fully retracted into the bushing and underground piping.

[0075] FIG. 29 shows a perspective view of aspects of an example drain riser assembly wherein the snap cap and main cap are not attached to the cylindrical member.

[0076] FIG. 30 shows a perspective view of aspects of the drain riser assembly of FIG. 29 wherein the snap cap and main cap are partially attached to the cylindrical member.

[0077] FIG. **31** shows a perspective view of aspects of the drain riser assembly of FIG. **29** wherein the snap cap and main cap are fully attached to the cylindrical member.

[0078] FIG. **32** shows a block diagram of a telescoping riser system according to some aspects of the disclosure.

[0079] FIG. **33** shows a perspective view of an example agricultural field that includes at least one telescoping riser assembly and at least one raised area.

[0080] FIG. **34** shows a perspective view of another portion of the agricultural field of FIG. **33**.

[0081] An artisan of ordinary skill in the art need not view, within isolated figure(s), the near infinite number of distinct permutations of features described in the following detailed description to facilitate an understanding of the present disclosure.

DETAILED DESCRIPTION

[0082] The present disclosure is not to be limited to that described herein. Mechanical, electrical, chemical, procedural, and/or other changes can be made without departing from the spirit and scope of the present disclosure. No features shown or described are essential to permit basic operation of the present disclosure unless otherwise indicated.

[0083] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which embodiments of the present disclosure pertain.

[0084] The terms “a,” “an,” and “the” include both singular and plural referents.

[0085] The term “or” is synonymous with “and/or” and means any one member or combination of members of a particular list.

[0086] The terms “invention” or “present invention” are not intended to refer to any single embodiment of the particular invention but encompass all possible embodiments as described in the specification and the claims.

[0087] The term “about” as used herein refers to slight variations in numerical quantities with respect to any quantifiable variable. Inadvertent error can occur, for example, through use of typical measuring techniques or equipment or from differences in the manufacture, source, or purity of components.

[0088] The term “substantially” refers to a great or significant extent. “Substantially” can thus refer to a plurality, majority, and/or a supermajority of said quantifiable variable, given proper context.

[0089] The term “generally” encompasses both “about” and “substantially.”

[0090] The term “configured” describes structure capable of performing a task or adopting a particular configuration. The term “configured” can be used interchangeably with other similar phrases, such as “constructed”, “arranged”, “adapted”, “manufactured”, and the like.

[0091] Terms characterizing sequential order, a position, and/or an orientation are not limiting and are only referenced according to the views presented.

[0092] The “scope” of the present disclosure is defined by the appended claims, along with the full scope of equivalents to which such claims are entitled. The scope of the disclosure is further qualified as including any possible modification to any of the aspects and/or embodiments disclosed herein which would result in other embodiments, combinations, subcombinations, or the like that would be obvious to those skilled in the art.

[0093] The term “agricultural equipment” encompasses any type of machinery associated with the agriculture industry. For example, both agricultural vehicles and agricultural implements are encompassed by the term “agricultural equipment”.

[0094] The term “agricultural activity” encompasses any type of agricultural operation performed in an agricultural field including, but not limited to, planting, harvesting, fertilizer application, pesticide application, herbicide application, plowing, ripping, discing, drilling, and the like. Agricultural activities can be performed with or without agricultural equipment.

[0095] As used herein, the term “exemplary” refers to an example, an instance, or an illustration, and does not indicate a most preferred embodiment unless otherwise stated.

[0096] Referring now to the figures, various aspects of an example embodiment of a drain riser assembly **10** are shown in FIGS. **1-4**. FIG. **1** shows a front elevation view of a drain riser assembly **10** according to some embodiments. The drain riser assembly **10** can also be referred to herein as the telescoping riser assembly **10**. Thus, “drain riser assembly **10**” and “telescoping riser assembly **10**” can be used interchangeably. As shown in FIG. **1**, according to some embodiments, the drain riser assembly **10** can comprise a cylindrical member **12**, a vibration isolator **14**, a snap cap **16**, and a main cap **18**. The cylindrical member **12** can include a first end **13** and a second end **15**. The cylindrical member can include one or more apertures **32**. The aperture(s) **32** can be generally circular according to some embodiments, however, they could be any suitable shape. The apertures **32** can number from one to N where N is any number greater than one. It should be noted that while the example cylindrical member **12** shown in FIGS. **1-4** includes many apertures **32**, only one aperture is labeled in FIGS. **1** and **2** for the sake of clarity. The snap cap **16** is removably attachable to one end of the cylindrical member **12**. More detail regarding such attachment is included below. The main cap **18** is removably attachable to the snap cap **16**, such that when the main cap **18** is attached to the snap cap **16** and the snap cap **16** is fully attached to the cylindrical member **12**, the main cap **18** forms a protective cap on one end of the cylindrical member **12**. The cylindrical member **12** can be inserted into, and slidably engage, the vibration isolator **14**. According to some embodiments, the telescoping riser assembly **10** further comprises underground piping **94**. According to some embodiments, the vibration isolator **14** can be and/or comprise a bushing **17** as is shown in FIGS. **1** and **2**. According to some embodiments, the vibration isolator **14** can be and/or comprise one or more tabs **99** integrally formed as part of the underground piping **94**. More detail regarding the sliding connection between the cylindrical member **12** and the vibration isolator **14** is included below.

[0097] Further, as shown in FIG. **1**, according to some embodiments, the cylindrical member **12** can be divided into three sections comprising a first section **24**, a second section **26**, and a third section **28**. According to some embodiments, the first and second sections **24**, **26** can be divided by a first cut line **20** and the second and third sections **26**, **28** can be divided by a second cut line **22**. The cylindrical member **12** can include any number of sections and/or any number of cut lines separating said sections according to some embodiments. Additionally, according to some embodiments, the cylindrical member **12** can comprise a single section with no cut lines.

[0098] FIG. **2** shows a side elevation view of the drain riser assembly **10** shown in FIG. **1**, wherein the snap cap **16** is not fully attached to the cylindrical member **12**. As shown in FIG. **2**, according to some embodiments, the cylindrical member **12** can include opposing guide slots **44**. Such guide slots **44** can span the length of the cylindrical member **12**. Such guide slots **44** can be, resemble, and/or comprise a trough, ditch, U-shaped bend, and/or similar structure. While only one guide slot **44** is visible in FIG. **2**, an identical guide slot **44** can be positioned on the opposite side of the cylindrical member **12**.

[0099] FIG. **3** shows a zoomed-in portion of the drain riser assembly **10** shown in FIG. **2**. FIG. **3** shows a view of the snap cap **16** interacting with the cylindrical member **12** to achieve attachment of the snap cap **16** to the cylindrical member **12**. According to some embodiments, the snap cap **16** comprises one or more protrusions **30** wherein each of the one or more protrusions **30** are configured to fit into and/or snap into an aperture **32** of the cylindrical member **12** in order to achieve attachment of the snap cap **16** to the cylindrical member **12**. According to some embodiments, each protrusion **30** can be press fit into an aperture **32**. While some embodiments include the snap cap **16** comprising two opposing protrusions **30**, each of which are configured to fit into and/or snap into an aperture **32** of the cylindrical member **12**, the snap cap **16** could have any number of protrusions ranging from one to N where N is any number greater than one. According to some embodiments, other types of connection means other than a protrusion fitting into and/or snapping into an aperture could be used, such connection means can include, but not limited to, screw(s), nut(s) and bolt(s), a threaded connection, other types of snap fit connection,

and the like.

[0100] FIG. 4 shows a zoomed-in portion of the drain riser assembly **10** shown in FIG. 2. FIG. 4 shows a view of the snap cap **16** interacting with the main cap **18** to achieve attachment of the snap cap **16** to the main cap **18**. According to some embodiments, the snap cap **16** can include an overmold **34** that molds around aspect(s) of the main cap **18** in order to achieve attachment of the main cap **18** to the snap cap **16**. As shown in FIG. 4, according to some embodiments, the main cap **18** can include a connection member **36**, wherein the connection member **36** is configured to interact with the overmold **34** of the snap cap **16** to facilitate attachment and/or connection of the snap cap **16** and main cap **18**. The overmold **34** and connection member **36** are described in greater detail below.

[0101] According to some embodiments, the cylindrical member **12**, vibration isolator **14**, snap cap **16**, and main cap **18** can be integrally formed such that all aspects of the drain riser assembly **10** shown in FIGS. 1 and 2 are formed as a single structure. According to some embodiments, the cylindrical diameter of any such components including the cylindrical member **12**, vibration isolator **14**, snap cap **16**, and/or main cap **18** can be any suitable length. For example, according to various embodiments, the cylindrical diameter of any of these components including the cylindrical member **12**, vibration isolator **14**, snap cap **16**, and/or main cap **18** could be 4, 6, 8, 10, or 12 inches. Additionally, if aspects of the drain riser assembly **10** are integrally formed as a single structure, according to some embodiments, the cylindrical diameter thereof can be any suitable length. For example, according to various embodiments, the cylindrical diameter of the integrally formed aspects of the drain riser assembly **10** could be 4, 6, 8, 10, or 12 inches.

[0102] According to some embodiments, the cylindrical member **12**, snap cap **16**, and main cap **18** can be integrally formed as a single structure, and the vibration isolator **14** can be integrally formed with underground piping **94** as another structure. The vibration isolator **14** can be and/or comprise one or more tabs **99** integrally formed with the underground piping **94**.

[0103] Additionally, according to some embodiments, the cylindrical member **12**, snap cap **16**, and main cap **18** can be integrally formed, and the vibration isolator **14** can be and/or comprise a bushing **17**, as described herein, wherein said bushing **17** is configured to interact with the underground piping **94** and the cylindrical member **12**.

[0104] FIGS. 5-8 show various views and various aspect(s) of an example embodiment of a cylindrical member **12**. The example embodiment of the cylindrical member **12** shown in FIGS. 5-8 could be used as the cylindrical member of the drain riser assembly **10** shown in FIGS. 1 and 2, according to some embodiments. FIG. 5 shows a side elevation view of the example embodiment of a cylindrical member **12**. The example embodiment of the cylindrical member **12** shown in FIG. 5 is configured to be telescopically extendable from and retractable into underground piping **94**. The underground piping **94** can be included as part of the drain riser assembly **10** according to some embodiments. Additionally, the term underground piping **94** can also be referred to herein as bottom riser **94** such that the terms “underground piping **94**” and “bottom riser **94**” can be used interchangeably. Further, the term cylindrical member **12** can be referred to herein as top riser **12** such that the terms “cylindrical member **12**” and “top riser **12**” can be used interchangeably. Such underground piping **94** is generally similar and/or identical to the cylindrical member **12** according to some embodiments. According to some embodiments, the cylindrical member **12** and underground piping **94** can be generally vertically oriented with relation to the ground. Thus, the underground piping **94** can extend in a generally downward direction from the surface of the ground at a vertical orientation wherein one end of the underground piping **94** is positioned at the surface of the ground and is exposed such that said end can receive the bushing **17** and/or cylindrical member **12** and the other end of the underground piping **94** is underground. While the cylindrical member **12** and underground piping **94** are vertically oriented according to some embodiments, the cylindrical member **12** and underground **94** need not be vertically oriented according to all embodiments. According to some embodiments, the diameter of the underground

piping **94** can be similar to, the same as, or slightly larger than the diameter of the cylindrical member **12** such that the cylindrical member **12** can extend from and/or retract into the underground piping **94** in a telescopic manner.

[0105] The height **38** of the first section **24** can be 12 inches according to some embodiments. The combined height **40** of the first section **24** and the second section **26** can be 24 inches according to some embodiments. The combined height **42** of the first, second, and third sections **24**, **26**, and **28** can be 36 inches according to some embodiments. Thus, according to some embodiments, each section of the cylindrical member **12** can be 12 inches in length. However, according to some embodiments, each section of the cylindrical member **12** could be any suitable length and/or the cylindrical member **12** itself could be any suitable length. Additionally, according to some embodiments, the diameter of each aperture **32** can be 1 inch. However, any suitably sized aperture could be included.

[0106] FIG. **5** further shows that the cylindrical member **12** can include a locking mechanism **46**. The locking mechanism **46** can comprise a single locking mechanism or can include opposing locking mechanisms **46** located on opposite sides of the cylindrical member **12**. Each locking mechanism **46** can be associated with a guide slot **44** of the cylindrical member **12**. According to some embodiments, the locking mechanism **46** can be, resemble, and/or comprise a cutout and/or slot of some sort wherein a tab of the vibration isolator **14**, such as a tab **53** of the bushing **17**, is configured to be inserted into and captured in each locking mechanism **46** in order to lock and/or secure the cylindrical member **12** in place relative to the vibration isolator **14**, wherein the vibration isolator **14** could be a bushing **17** according to some embodiments.

[0107] A user can twist and/or spin the cylindrical member **12** radially relative to the vibration isolator **14** and/or manipulate the cylindrical member in a vertical direction in order to insert and/or capture a tab of the vibration isolator **14**, such as a tab **53** of the bushing **17**, in each locking mechanism **46**. When the cylindrical member **12** is locked and/or secured in place relative to the vibration isolator **14**, the cylindrical member **12** is not able to move in terms of extension and/or retraction relative to the underground piping **94** and/or the vibration isolator **14**. A user can twist, spin, and/or manipulate the cylindrical member **12** to release the tab of the vibration isolator **14**, such as a tab **53** of the bushing **17**, from each locking mechanism **46** such that the cylindrical member **12** is free, unlocked, and movable relative to the vibration isolator **14** and the underground piping **94**. When the cylindrical member **12** is unlocked, the cylindrical member **12** is movable and/or slidable relative to the vibration isolator **14** and the underground piping **94** such that the cylindrical member **12** can telescopically extend from and retract into the vibration isolator **14** and/or underground piping **94**. The cylindrical member **12** is configured to slidably engage the vibration isolator **14** and/or the underground piping **94**.

[0108] FIG. **6** provides a zoomed-in view of the locking mechanism **46** shown in FIG. **5**.

[0109] FIG. **7** shows a top plan view of the cylindrical member **12**. The U-shaped nature of each guide slot **44** can be seen in FIG. **7**. As discussed in further detail below, each guide slot **44** can interact with a tab of the vibration isolator **14**, such as a tab **53** of the bushing **17**, to facilitate sliding the cylindrical member **12** relative to the vibration isolator **14**. According to some embodiments, the diameter **47** of the cylindrical member **12** can be 4.957 inches. However, the diameter **47** of the cylindrical member **12** can be any suitable length. For example, according to various embodiments, the diameter **47** of the cylindrical member **12** could be 4, 5, 6, 8, 10, or 12 inches.

[0110] FIG. **8** shows a zoomed-in view of aspects of the cylindrical member **12** of FIG. **7** including a guide slot **44**. As shown in FIG. **8**, the diameter **49** of the guide slot **44** can be 0.790 inches according to some embodiments. However, the diameter **49** of the guide slot **44** can be any suitable length. FIG. **8** shows that each guide slot **44** comprises a back **48** according to some embodiments. According to some embodiments, the back **48** of each guide slot **44** can be used to facilitate attachment of the snap cap **16** and/or main cap **18** to the cylindrical member **12**. The back **48** of

each guide slot **44** can facilitate sliding engagement of the snap cap **16** with the cylindrical member **12**. Further detail regarding interaction between the snap cap **16** and the cylindrical member **12** is provided below.

[0111] FIGS. **9-11** show various views of an example embodiment of a bushing **17**. As noted, according to some embodiments, the vibration isolator **14** can be and/or comprise a bushing **17**. The example embodiment of the bushing **17** shown in FIGS. **9-11** could be used as the vibration isolator and/or bushing of the drain riser assembly **10** shown in FIGS. **1** and **2**, according to some embodiments. FIG. **9** shows a perspective view of the example embodiment of a bushing **17**. As shown in FIG. **9**, the bushing **17** can include a bushing body **50** wherein the body has a first end **51**, a second end **52**, an inner surface **55**, and an outer surface **57**; one or more tabs **53** (as noted above); one or more flanges **54**; a rib **56**; and a rim **58**.

[0112] The body **50** of the bushing **17** can be generally cylindrical such that it can receive the cylindrical member **12** via its inner surface **55**. The first and second ends **51**, **52** of the bushing can be open such that the cylindrical member **12** can slidingly pass through the interior of the bushing **17**.

[0113] As noted, while the bushing **17** can include any number of tabs **53**, as an example, the embodiment shown in FIGS. **9-11** includes two opposing tabs **53**. Each tab **53** can be any sort of protrusion, knob, projection, or any similar structure. Each tab **53** can extend radially inwardly from the inner surface **55** of the bushing **17**. Each tab **53** can be configured as four pieces, and/or “ears”, forming a “cross” shape as shown in the example embodiment of FIG. **9**. As noted, the tabs **53** can interact with the guide slots **44** of the cylindrical member **12** to facilitate the sliding nature of the cylindrical member **12** relative to the bushing **17**, which allows for the cylindrical member **12** to selectively (based on user input) telescopically extend from and retract into the bushing **17** and/or the underground piping **94**. According to some embodiments, the tabs **53** can be configured to slidingly engage the guide slots **44**. The tabs **53**, and/or ears thereof, can fit into the guide slots **44** according to some embodiments. According to some embodiments, the bushing has opposing tabs **53** that align with opposing guide slots **44** of the cylindrical member **12**. This alignment facilitates easy and effective sliding of the cylindrical member **12** relative to the bushing **17**. Additionally, each tab **53** is configured to be able to be inserted into a locking mechanism **46** of the cylindrical member **12**. According to some embodiments, the bushing **17** has opposing tabs **53** that align with opposing locking mechanisms **46** of the cylindrical member **12**. According to some embodiments, when viewing the tabs **53** from above, such as in FIG. **10**, the diameter of the circular portion of each tab **53** can be 0.75 inches. However, the diameter of the circular portion of each tab **53** can be any suitable size.

[0114] Each flange **54** can be a protrusion, projection, and/or other similar structure that extends radially outwardly from the outer surface **57** of the bushing **17**. Additionally, each flange **54** can extend along the length **62** of the bushing **17** as shown in FIGS. **9** and **11**. The bushing **17** can be inserted into the exposed end of the underground piping **94**. Also, the bushing **17** is removable from the underground piping **94**. The flange(s) **54** are configured to provide for proper fitting of the bushing **17** and also to act as a stabilizer and/or buffer to stabilize the bushing **17** and prevent and/or mitigate any unwanted movement of the bushing **17** when the bushing **17** is inserted into the underground piping **94**. According to some embodiments, the bushing **17** can be inserted into the underground piping **94** wherein the bushing **17** contacts the underground piping via the outer surface **57** and/or the flange(s) **54**. The flange(s) **54** also provide spacing such that the rib **56** can rest on an outer edge of each flange **54** wherein space exists between the rib **56** and the outer surface **57** of the body **50** of the bushing **17** as shown in FIG. **9**. While the example bushing of FIGS. **9-11** shows eight flanges **54**, any number of flanges ranging from one to N where N is any number greater than one could be included. It should be noted that while the example bushing **17** shown in FIGS. **9-11** includes eight flanges **54**, only some flanges are labeled in FIGS. **9-11** for the sake of clarity.

[0115] The bushing **17** can include a rib **56** as shown in FIG. **9**. The rib **56** can extend radially around the outer surface **57** of the bushing **17** at and/or near one end of the bushing **17**. For example, the embodiment of FIGS. **9-11** shows the rib **56** extending radially around the outer surface **57** of the bushing **17** at and/or near the first end **51** of the bushing **17**. The rib **56** can be connected to the bushing **17** via an outer edge of each flange **54** such that space exists between the rib **56** and the outer surface **57** of the body **50**. While not shown in the embodiment of FIGS. **9-11**, according to some embodiments, screws and/or other attachment means can be used in conjunction with the rib **56** to facilitate secure attachment of the bushing **17** to the underground piping **94**. For example, screw(s) can be inserted through the rib **56** and the underground piping **94** to create an effective attachment between the bushing **17** and the underground piping **94**. Thus, according to some embodiments, the rib **56** can comprise one or more apertures for which screw(s) can be inserted.

[0116] The bushing **17** can include a rim **58**. According to some embodiments, the rim **58** can be a generally flat, circular structure and can be positioned at and/or near one end of the bushing **17**. For example, in the embodiment of FIGS. **9-11**, the rim **58** is positioned at the first end **51** of the body **50** of the bushing **17**. The rim **58** can be in contact with the rib **56** as is shown in FIG. **9**. The rim **58** can extend radially outwardly from the body **50**. When the bushing **17** is fully inserted into the underground piping **94**, the rim **58** can be flush with and/or rest on the surface of the ground.

[0117] FIG. **10** shows a top plan view of the bushing **17**. According to some embodiments, the total diameter **60** of the bushing **17**, inclusive of the rim **58**, can be 6.755 inches. However, the diameter **60** of the bushing **17** can be any suitable size. According to various embodiments, the diameter of the bushing **17**, inclusive or exclusive of the rim **58**, can be any suitable size. For example, according to various embodiments, the diameter of the bushing **17**, inclusive or exclusive of the rim **58**, could be 4, 6, 8, 10, or 12 inches.

[0118] FIG. **11** shows a side elevation view of the bushing **17**. According to some embodiments, the total length **62** of the bushing **17** can be 4.5 inches. However, the length **62** of the bushing **17** can be any suitable length.

[0119] FIGS. **12** and **13A** show various views of underground piping **94** according to some embodiments. FIG. **12** shows a side elevation view of underground piping **94**. As shown in FIG. **12**, according to some embodiments, the underground piping **94** can comprise a top **96**, a bottom **97**, zero or more aperture(s) **100**, zero or more large separator(s) **102**, zero or more medium separator(s) **104**, zero or more small separator(s) **106**, and an outer wall **110**.

[0120] The underground piping **94** can be made of any suitable material including, but not limited to, plastic, metal, and the like. As shown in FIGS. **12** and **13A**, the underground piping **94** can be generally cylindrical and/or tube-like. According to some embodiments, the underground piping **94** can be positioned generally vertically in the ground in and/or near an agricultural field wherein the top **96** of the underground piping is level with, flush with, at, and/or near ground level such that the top **96** of the underground piping **94** is generally exposed to the outside air above ground.

Additionally, the top **96** of the underground piping **94** can be open such that a cylindrical member **12** can slidably engage the underground piping **94** wherein the cylindrical member **12** can be retracted into and extended from the underground piping **94** in a telescopic manner.

[0121] As shown in FIG. **12**, the underground piping **94** can include a bottom **97**. According to some embodiments, the bottom **97** can be open such that the underground piping **94** can be operatively connected to additional piping and/or other aspects of a drainage system that are positioned below ground in order to drain excess water and/or control erosion.

[0122] According to some embodiments, the underground piping **94** can include zero or more apertures **100**. While the embodiment shown in FIG. **12** shows the underground piping **94** including many apertures **100**, only one is labeled for the sake of clarity. According to some embodiments, the number of apertures **100** can range from zero to N where N is any number greater than zero. The apertures **100** can be generally circular in shape, as is shown in FIG. **12**, but

the apertures **100** could be any suitable shape according to various embodiments. The apertures **100** of the underground piping **94** can be smaller in diameter than the apertures **32** of the cylindrical member **12** according to some embodiments. The smaller diameter of the apertures **100** of the underground piping **94** allow for ground water to enter the underground piping **94** while keeping soil and debris out. Letting ground water into the underground piping **94** while keeping soil and other types of debris out allows the underground piping **94** to function properly to drain excess water and/or control erosion without being slowed and/or rendered less effective due to clogging, damage, and/or other issues caused by the buildup of soil and/or other debris.

[0123] As shown in FIG. **12**, the underground piping **94** can be divided into sections that are divided by separators. Multiple types of separators can be used including large separator(s) **102**, medium separator(s) **104**, and small separator(s) **106**. While FIG. **12** shows that the underground piping **94** includes several of each kind of separator **102**, **104**, **106**, the underground piping **94** can include any number of separators **102**, **104**, **106** and sections. Only one of each type of separator is labeled in FIG. **12** for the sake of clarity. Dividing the underground piping **94** into sections via the use of separator(s) **102**, **104**, **106** can contribute to the structural integrity of the underground piping **94**.

[0124] The underground piping **94** can be of any suitable length **95**. For example, the underground piping could be 1 foot, 2 feet, 3 feet, 4 feet, and/or any other suitable length. The example underground piping **94** shown in FIG. **12** has a length **95** of 46 inches.

[0125] FIG. **13A** shows a top plan view of the underground piping **94** of FIG. **12**. The underground piping **94** of FIGS. **12** and **13A** is an embodiment wherein the vibration isolator **14** is integrally formed as part of the underground piping **94**. Thus, the embodiment of the underground piping **94** of FIGS. **12** and **13A** does not require the use of a bushing **17**. The vibration isolator **14** can include one or more tabs **99** integrally formed as part of the underground piping **94** on an inner wall **108** of the underground piping **94** such that the tab(s) **99** extend inwardly toward the center of the underground piping **94** as shown in FIG. **13A**. For example, the embodiment of FIGS. **12** and **13A** shows the vibration isolator **14** including two tabs **99**. Just like the bushing **17** and tab(s) **53** thereof, the one or more tabs **99** are configured to align with and slidably engage the guide slot(s) **44** of the cylindrical member **12** and/or align with and interact with the locking mechanism(s) **46** of the cylindrical member **12**. Thus, the tab(s) **99** properly align the cylindrical member **12** and the underground piping **94**. The tab(s) **99** of the integrally formed vibration isolator **14** are configured to function in the same and/or similar manner as the tab(s) **53** of the bushing **17** according to some embodiments. Additionally, the tab(s) **99** of the integrally formed vibration isolator **14** are configured to have the same and/or similar properties/characteristics/attributes as the tab(s) **53** of the bushing **17** according to some embodiments. Additionally, according to embodiments in which the vibration isolator **14** comprises tab(s) **99** integrally formed with the underground piping **94** rather than comprising a bushing **17**, such tab(s) **99** can be positioned in generally the same or similar location relative to the top **96** of the underground piping **94** as the tab(s) **53** of the bushing **17** in embodiments wherein the vibration isolator **14** comprises a bushing **17**. For example, according to some embodiments, the tab(s) **99** can be positioned on the inner wall **108** of the underground piping **94** six inches or less from the top **96** of the underground piping **94**.

[0126] FIG. **13A** also shows a diameter **98** of the underground piping **94**. According to various embodiments, the diameter **98** can be any suitable length such that the cylindrical member **12** can slide into the underground piping **94**. For example, according to some embodiments, the diameter **98** can be 6.45 inches. Additionally, the diameter **98** of the underground piping **94** could be 4, 6, 8, 10, or 12 inches according to various embodiments.

[0127] FIG. **13B** shows a perspective view generally of the top **96** of underground piping **94** according to some embodiments. The underground piping **94** of FIG. **13B** is shown above ground prior to below ground installation. The underground piping of FIG. **13B** includes a vibration isolator **14** positioned on the inner wall **108** of the underground piping **94** wherein the vibration

isolator **14** comprises two tabs **99** integrally formed with the underground piping **94**. The example underground piping **94** of FIG. **13B** is shorter in length than many embodiments of the underground piping **94**. FIG. **13B** is included as an example view of particular aspects of the underground piping **94**.

[0128] All aspects of the underground piping **94** shown in FIG. **12** can be included in various embodiments thereof whether the vibration isolator **14** is a removable bushing **17** and/or is integrally formed with the underground piping **94**.

[0129] Although not required, embodiments of the telescoping riser assembly **10** that are used in conjunction with already existing tile draining systems wherein the underground piping is already installed, can utilize a bushing **17** as the vibration isolator **14**. Although not required, embodiments of the telescoping riser assembly **10** that are used in conjunction with new tile draining systems to be installed in an agricultural field, can utilize underground piping **94** wherein the vibration isolator **14** is and/or comprises one or more tab(s) **99** integrally formed with the underground piping **94**. Thus, by installing underground piping **94** wherein the vibration isolator **14** is and/or comprises one or more tabs **99** integrally formed with the underground piping **94**, the new tile drainage system to be installed can eliminate the need to utilize one or more bushings **17**.

[0130] FIGS. **14-17** show various views of an example embodiment of a snap cap **16**. The example embodiment of the snap cap **16** shown in FIGS. **14-17** could be used as the snap cap of the drain riser assembly **10** shown in FIGS. **1** and **2**, according to some embodiments. FIG. **14** shows a perspective view of the example embodiment of a snap cap **16**. According to the embodiment of FIGS. **14-17**, the snap cap **16** can comprise side wall(s) **64**, one or more slots **70**, one or more protrusions **30**, a top end **66**, a bottom end **68**, and an overmold **34**.

[0131] As seen in FIGS. **14-17**, the snap cap **16** can be generally cylindrical according to some embodiments and can have side walls **64** that are curved. The snap cap **16** is configured to be attachable to one end of the cylindrical member **12**. According to some embodiments, attachment of the snap cap **16** to the cylindrical member **12** involves sliding the snap cap **16** onto an end of the cylindrical member **12**. The slots **70** of the snap cap help facilitate such sliding. While the embodiment of FIGS. **14-17** shows two opposing slots **70**, any number of slots **70** could be included. The opposing slots **70** of the embodiment of FIGS. **14-17** are configured to align with the back **48** of each opposing guide slot **44** of the cylindrical member **12**. In this fashion, the snap cap **16** is configured to be slid onto an end of the cylindrical member **12** wherein the slots **70** of the snap cap **16** at least partially surround the back **48** of each guide slot **44** of the cylindrical member **12**. Therefore, the guide slots **44**, and the backs **48** thereof, of the cylindrical member **12** and the slots **70** of the snap cap **16** slidingly interact to facilitate sliding engagement between and attachment of the snap cap **16** and the cylindrical member **12**.

[0132] The fact that the slots **70** of the snap cap **16** fit into the guide slots **44** of the cylindrical member **12**, or vice versa, also serves to align the protrusion(s) **30** of the snap cap **16** with aperture(s) **32** of the cylindrical member **12**. As noted above with reference to FIG. **3**, the protrusion(s) **30** are each configured to fit into and/or snap into an aperture **32** of the cylindrical member **12** to secure the snap cap **16** to the cylindrical member **12**. Each protrusion **30** can be generally circular and can extend outwardly from the side wall(s) **64** of the snap cap **16**. A user can press fit the protrusion(s) **30** into aperture(s) **32** in order to secure and/or attach the snap cap **16** to the cylindrical member **12**. A user can remove the protrusion(s) **30** from the aperture(s) **32** in order to unsecure the snap cap **16** from the cylindrical member **12** to remove the snap cap **16** from the cylindrical member **12**. The embodiment of FIGS. **14-17** includes two opposing protrusions **30** positioned on opposite sides of the snap cap **16**, however, any suitable number of protrusions **30** could be included in any suitable formation.

[0133] FIG. **15** shows a top plan view of the snap cap **16**. As shown in FIGS. **14** and **15**, the top end **66** of the snap cap **16** is closed and the bottom end **68** is open such that the snap cap **16** forms a generally cylindrical shape having a closed top, side walls, and an open bottom. The top end **66** of

the snap cap **16** can comprise an overmold **34** that extends upward from the top end **66**. The overmold **34** can be generally circular in shape and can be positioned generally at or near the center of the top end **66** of the snap cap **16**. According to some embodiments, the diameter **71** of the snap cap **16** from side wall **64** to side wall **64** can be 4.69 inches, however, the diameter **71** could be any suitable size. For example, according to various embodiments, the diameter **71** could be 4, 6, 8, 10, or 12 inches.

[0134] FIG. **16** shows a side elevation view of the snap cap **16** and FIG. **17** shows a front elevation view of the snap cap **16**. As shown in FIGS. **16** and **17**, the overmold **34** can be narrower where it contacts the top end **66** of the snap cap **16** and widen as it extends away from the top end **66**. This variance in the width of the overmold **34** helps to facilitate attachment of the main cap **18** to the snap cap **16**. According to some embodiments, the height **72** of the side walls **64** of the snap cap **16** can be 2.585 inches and the total height **74** of the snap cap **16**, inclusive of the overmold **34**, can be 2.745 inches. However, any suitable sizes can be used for the height **72** of the side walls **64** and/or the total height **74** of the snap cap **16**.

[0135] FIGS. **18-20** show various views of the main cap **18**. The example embodiment of the main cap **18** shown in FIGS. **18-20** could be used as the main cap of the drain riser assembly **10** shown in FIGS. **1** and **2**, according to some embodiments. FIG. **18** shows a perspective view of the main cap **18**. As shown in FIGS. **18-20**, the main cap **18** can include a main cap body **76** (wherein the body **76** has a top **78** and a bottom **80**) and a connection member **36** wherein the connection member **36** has a raised portion **82**, a central portion **84**, and one or more legs **86** wherein each leg has an end member **87**.

[0136] The main cap **18** is configured to attach to the snap cap **16** wherein the overmold **34** of the snap cap **16** and the connection member **36** of the main cap **18** interact to facilitate the attachment. Thus, when the main cap **18** is attached to the snap cap **16** and the snap cap **16** is attached to the cylindrical member **12**, the main cap **18** covers one of the ends of the cylindrical member **12**.

[0137] As shown in FIGS. **18-20**, the main cap body **76** is generally circular and can be disc-like according to some embodiments. According to various embodiments, the main cap body **76** can be made of and/or comprise plastic, metal (such as steel), polycarbonate, any other suitable material, and/or any combination thereof. The connection member **36** can be located generally in the center of main cap body **76** and extend upward from the top **78** of the main cap body **76**. The connection member **36** can comprise a raised portion **82** and a plate **84**. The raised portion **82** and/or plate **84** can be cored and/or hollowed out to allow space for the overmold **34** of the snap cap **16** such that the connection member **36** can matingly engage the overmold **34** to secure the main cap **18** to the snap cap **16**. While FIGS. **18-20** show the plate **84** having four legs **86**, according to some embodiments the plate **84** can have one or more legs **86** extending from it generally laterally and/or wherein each leg **86** can extend in a downward direction relative to the plate **84** toward the outward end of each leg **86**. The end of each leg **86** can include an end member **87** that is generally circular in shape.

[0138] According to some embodiments, the diameter **88** of the body **76** of the main cap **18** can be 7.89 inches. However, the diameter **88** can be any suitable size. For example, according to various embodiments, the diameter **88** of the main cap **18** can be 4 inches, 6 inches, 8 inches, 10 inches, 12 inches, and/or any length greater than 12 inches. According to some embodiments, the diameter **90** of the plate **84** of the connection member **34** can be 1 inch. However, the diameter **90** can be any suitable size. According to some embodiments, the height **92** of the body **76** of the main cap **18** can be 0.245 inches. However, the height **92** could be any suitable size.

[0139] FIGS. **21-24** show various views of a bushing **17** and underground piping **94**. FIG. **21** shows a perspective view of the bushing **17** and underground piping **94** wherein the bushing **17** is not attached to the underground piping **94**. FIGS. **22** and **23** show two different perspective views of the bushing **17** and underground piping **94** wherein the bushing **17** is partially inserted into the underground piping **94**. FIG. **24** shows a perspective view of the bushing **17** and underground

piping **94** wherein the bushing **17** is fully inserted into and attached to the underground piping **94**. [0140] FIGS. **21-24** show that the underground piping **94** includes an exposed open end that is positioned at or near the surface of the ground, wherein said exposed end is shown to be the top **96** of the underground piping **94** in FIG. **21**. The progression of FIGS. **21-24** show how the bushing **17** can be inserted into and attached to the exposed open end of the underground piping **94**. The flanges **54** of the bushing **17** can contact the inner wall **108** of the underground piping **94** to help secure and/or attach the bushing **17** to the underground piping **94**. When the bushing **17** is fully inserted into and secured to the underground piping **94**, the rim **58** of the bushing **17** is positioned at or near the surface of the ground.

[0141] FIGS. **25-28** show various views of a cylindrical member **12**, bushing **17**, and underground piping **94**, wherein a snap cap **16** and main cap **18** are attached to the cylindrical member **12**. FIG. **25** shows a perspective view of a farmer preparing to insert the cylindrical member **12** into the bushing **17**.

[0142] FIG. **26** shows the cylindrical member inserted into the bushing **17** and locked in place relative to the bushing **17** and the underground piping **94** via the locking mechanism(s) **46** and/or tab(s) **53**. The locked position of the cylindrical member **12** wherein the cylindrical member **12** extends above the surface of the ground as shown in FIG. **26** represents the “normal” position of the cylindrical member **12**. In other words, whenever a farmer is not performing an agricultural activity (such as planting, harvesting, and the like) in the agricultural field, the locked and extended position of the cylindrical member **12** as shown in FIG. **26** is the position that the cylindrical member **12** will maintain.

[0143] FIG. **27** shows a perspective view of the cylindrical member **12** wherein the cylindrical member has been unlocked and has been partially retracted into the bushing **17** and the underground piping **94**. As shown in FIG. **27**, when the cylindrical member is partially retracted, more of its length is underground as compared to when the cylindrical member **12** is fully extended and in the locked position. According to some embodiments, the telescopic extension and retraction of the cylindrical member **12** can be accomplished via sliding the cylindrical member **12** relative to the bushing **17** wherein the tab(s) **53** of the bushing **17** are within the guide slot(s) **44** of the cylindrical member **12**, which facilitates the sliding.

[0144] FIG. **28** shows a perspective view when the cylindrical member **12** is fully retracted into the bushing **17** and underground piping **94**. As shown in FIG. **28**, when the cylindrical member **12** is fully retracted, the main cap **18** is flush with the ground and/or fits firmly to the ground. The fully retracted position of the cylindrical member **12** shown in FIG. **28** is used when a farmer is performing an agricultural activity such as planting, harvesting, and the like. By retracting the cylindrical member **12** into the bushing **17** and underground piping **94** such that the cylindrical member **12** is underground and the main cap **18** is flush with the surface of the ground and/or fits firmly to the surface of the ground, a farmer can perform the agricultural activity in the field without needing to waste time, money, effort, and resources to navigate around each drain riser assembly **10**. A farmer can simply pass right over the main cap **18** when performing an agricultural activity. Additionally, since the main cap **18** is flush with the surface of the ground and/or fits firmly to the surface of the ground when the cylindrical member **12** is retracted, the main cap protects the drain riser assembly **10**, including the underground piping **94**, from being damaged. Also, since the main cap **18** is flush with the surface of the ground and/or fits firmly to the surface of the ground when the cylindrical member **12** is retracted, the main cap **18** prevents debris from entering the drain riser assembly **10**, including the bushing **17** and/or the underground piping **94**. The main cap **18** acts as a shield in this way to protect the drain riser assembly **10** from damage and/or debris.

[0145] It should be noted that while FIGS. **21-28** include the use of a bushing **17** as the vibration isolator **14**, tab(s) **99** integrally formed with the underground piping **94** could be used as the vibration isolator **14** instead of the bushing **17** according to various embodiments.

[0146] FIGS. **29-31** show various views of a cylindrical member **12**, snap cap **16**, and main cap **18**. FIG. **29** shows a perspective view of a farmer preparing to attach the snap cap **16**, and main cap **18**, to the cylindrical member **12**. FIG. **30** shows a perspective view of the snap cap **16**, and main cap **18**, in the process of slidingly being attached to the cylindrical body **12**. FIG. **31** shows a perspective view of the snap cap **16**, and main cap **18**, fully attached to the cylindrical body **12**. As shown in FIG. **31**, the body **76** of the main cap **18** rests at or near and/or is in contact with an end of the cylindrical member **12** when fully attached.

[0147] It is noted that any component(s) shown and/or described in FIGS. **21-31** could be included as part of the drain riser assembly **10** of FIGS. **1** and **2**, according to some embodiments.

[0148] It is noted that each component of the drain riser assembly **10** can be made of and/or comprise plastic, metal (such as steel), polycarbonate, any other suitable material, and/or any combination thereof. Additionally, each component of the drain riser assembly **10** is configured to be resistant to ultra-violet (UV) rays including, but not limited to, being resistant to both UVA and UVB rays.

[0149] FIG. **32** shows a block diagram of a telescoping riser system **200**. According to some embodiments, the system **200** can comprise one or more telescoping riser assemblies **202**, a human-machine interface (“HMI”) **204** (wherein the HMI **204** can comprise a user interface (“UI”) **206**), a positioning system **208** (wherein the positioning system **208** can comprise one or more electronic locators **209**), a controller/processing system **210**, component(s) **212** for communication, a network **213**, a memory **214**, and a database **216**.

[0150] The system **200** can be used to control and/or monitor one or more telescoping riser assemblies **202** in a portion of an agricultural field, in the entirety of an agricultural field, and/or across more than one agricultural field. The system **200** allows for a user to determine and/or monitor the position/location of each of the one or more telescoping riser assemblies **202** such that a user can pinpoint the exact location of each telescoping riser assembly that is part of the system **200**. By allowing a user to determine the exact location of each telescoping riser assembly that is part of the system **200**, a user can quickly, easily, and effectively extend and/or retract the top riser/cylindrical member at different times throughout the agricultural cycle.

[0151] The one or more telescoping riser assemblies **202** can comprise any number of telescoping riser assemblies ranging from one to N where N is any number greater than one. According to some embodiments, each of the one or more telescoping riser assemblies **202** can be the drain riser assembly/telescoping riser assembly **10** described herein.

[0152] The HMI **204** can be any suitable device capable of outputting information and/or communicating with other electronic and/or mechanical component(s). Examples of devices that could be used as the HMI **204** include, but are not limited to, a computer, a laptop, a tablet, a smartphone, a mobile device, a handheld device, and/or any other type of computing device or smart device. The HMI **204** can be operationally connected to each of the one or more telescoping riser assemblies **202**.

[0153] According to some embodiments, the HMI **204** can include a user interface (“UI”) **206**. A UI is how the user interacts with a machine. The UI **206** can be a digital interface, a command-line interface, a graphical user interface (“GUI”), oral interface, virtual reality interface, or any other way a user can interact with a machine (user-machine interface). For example, the UI **206** can include a combination of digital and analog input and/or output devices or any other type of UI input/output device required to achieve a desired level of control and/or monitoring of aspect(s) of the system **200**. Nonlimiting examples of input and/or output devices include computer mice, keyboards, touchscreens, knobs, dials, switches, buttons, speakers, microphones, printers, LIDAR, RADAR, etc. Input(s) received by the UI **206** can then be sent to the controller **210**, a microcontroller, and/or any other controller to control operational aspect(s) and/or monitor aspect(s) of the system **200**.

[0154] The UI **206** can include a display, which can act as an input and/or output device. More

particularly, the display can be a liquid crystal display (“LCD”), a light-emitting diode (“LED”) display, an organic LED (“OLED”) display, an electroluminescent display (“ELD”), a surface-conduction electron emitter display (“SED”), a field-emission display (“FED”), a thin-film transistor (“TFT”) LCD, a bistable cholesteric reflective display (i.e., e-paper), a touch-screen display, etc. The UI **206** also can be configured with a microcontroller and/or the controller **210** to display conditions or data associated with the system **200** in real-time or substantially real-time. [0155] The system **200** can comprise a positioning system **208**. The positioning system **208** could also be called a location determining system. The positioning system **208** can be used to determine and/or monitor the location of each of the one or more telescoping riser assemblies **202**. The positioning system **208** could utilize the global positioning system (“GPS”) and/or any other suitable positioning/location determination system. According to some embodiments, the positioning system **208** can comprise one or more electronic locator(s) **209** wherein one electronic locator **209** is operationally attached to each of the one or more telescoping riser assemblies **202**. The positioning system **208** can utilize GPS, and/or any other similar location determining system, in conjunction with the one or more electronic locators **209** to pinpoint, determine, and/or monitor the location of each of the one or more telescoping riser assemblies **202**. Each of the one or more electronic locators **209** can be battery-powered according to some embodiments. The positioning system **208** can be operationally connected to each of the one or more telescoping riser assemblies **202** and/or to the HMI **204**.

[0156] With respect to the battery and/or batteries used to power the one or more electronic locators **209**, a dry cell battery may be used. Additionally, the battery may be rechargeable, such as a lead-acid battery, a low self-discharge nickel metal hydride battery (“LSD-NiMH”), a nickel-cadmium battery (“NiCd”), a lithium-ion battery, or a lithium-ion polymer (“LiPo”) battery. Careful attention should be taken if using a lithium-ion battery or a LiPo battery to avoid the risk of unexpected ignition from the heat generated by the battery. While such incidents are rare, they can be minimized via appropriate design, installation, procedures, and layers of safeguards such that the risk is acceptable.

[0157] As mentioned, according to some embodiments, a satellite-based radio-navigation system such as GPS is used. GPS uses satellites to provide geolocation information to a GPS receiver. GPS, and other satellite-based radio-navigation systems, can be used for location positioning, navigation, tracking, and mapping.

[0158] The system **200** can include an intelligent control/controller **210**. Examples of such a controller may be processing units alone or other subcomponents of computing devices. The controller **210** can also include other components and can be implemented partially or entirely on a semiconductor (e.g., a field-programmable gate array (“FPGA”)) chip, such as a chip developed through a register transfer level (“RTL”) design process. According to some embodiments, the controller **210** can be part of the HMI **204**. According to some embodiments, the controller **210** is separate from the HMI **204**.

[0159] A processing unit, also called a processor, is an electronic circuit which performs operations on some external data source, usually memory or some other data stream. Non-limiting examples of processors include a microprocessor, a microcontroller, an arithmetic logic unit (“ALU”), and most notably, a central processing unit (“CPU”). A CPU, also called a central processor or main processor, is the electronic circuitry within a computer that carries out the instructions of a computer program by performing the basic arithmetic, logic, controlling, and input/output (“I/O”) operations specified by the instructions. Processing units are common in tablets, telephones, handheld devices, laptops, user displays, smart devices (TV, speaker, watch, etc.), and other computing devices.

[0160] The system **200** can further include component(s) **212** for establishing communications. Data and/or electronic communication can occur between any component(s) of the system **200** including the positioning system **208** which can include use of a GPS system (and/or similar

location determining system), the one or more electronic locator(s) **209** operationally attached to each of the one or more telescoping riser assemblies **202**, the HMI **204**, the controller **210**, the memory **214**, and/or the database **216**. According to some embodiments, such data and/or electronic communication can occur via the network **213**. The component(s) **212** for establishing communications can include any combination of modem(s), router(s), access point(s), bridge(s), gateway(s), hub(s), repeater(s), switch(es), transceiver(s), and the like in order to facilitate data and/or electronic communication. The communications component(s) **212** can be configured to perform data communication wirelessly and/or in a wired fashion. According to some embodiments, the communications component(s) can include one or more communications ports such as Ethernet, serial advanced technology attachment (“SATA”), universal serial bus (“USB”), or integrated drive electronics (“IDE”), for transferring, sending, receiving, and/or or storing data.

[0161] According to some embodiments, the communications component(s) **212** are able to perform data communication either within the system **200** and/or externally of the system **200** in a wireless fashion using any sort of wireless connection device and/or protocol. This can include, but is not limited to, Bluetooth, Wi-Fi, cellular data, radio waves, satellite, and/or generally any other form of wireless connection. Therefore, the communications component(s) **212** will include generally any electronic component(s) necessary to allow for such wireless communication.

[0162] According to some embodiments, the communications component(s) **212** are able to perform data and/or electronic communication either within the system **200** and/or externally of the system **200** via a wired connection. Wired communication can take the form of CAN bus, Ethernet, co-axial cable, fiber optic line, and/or can include generally any other device and/or protocol which will allow for wired communication. Therefore, the communications component(s) **212** will include generally any electronic component(s) necessary to allow for such wired communication.

[0163] Ethernet is a family of computer networking technologies commonly used in local area networks (“LAN”), metropolitan area networks (“MAN”) and wide area networks (“WAN”). Systems communicating over Ethernet divide a stream of data into shorter pieces called frames. Each frame contains source and destination addresses, and error-checking data so that damaged frames can be detected and discarded; most often, higher-layer protocols trigger retransmission of lost frames. As per the OSI model, Ethernet provides services up to and including the data link layer. Ethernet was first standardized under the Institute of Electrical and Electronics Engineers (“IEEE”) 802.3 working group/collection of IEEE standards produced by the working group defining the physical layer and data link layer's media access control (“MAC”) of wired Ethernet. Ethernet has since been refined to support higher bit rates, a greater number of nodes, and longer link distances, but retains much backward compatibility. Ethernet has industrial application and interworks well with Wi-Fi. The Internet Protocol (“IP”) is commonly carried over Ethernet and so it is considered one of the key technologies that make up the Internet.

[0164] According to some embodiments, data and/or electronic communication can occur over the network **213**. According to some embodiments, the network **213** is, by way of example only, a wide area network (“WAN”) such as a TCP/IP based network or a cellular network, a local area network (“LAN”), a neighborhood area network (“NAN”), a home area network (“HAN”), or a personal area network (“PAN”) employing any of a variety of communication protocols, such as Wi-Fi, Bluetooth, ZigBee, near field communication (“NFC”), etc., although other types of networks are possible and are contemplated herein. The network **213** typically allows communication between aspect(s) of the system **200** during moments of low-quality connections. Communications through the network **213** can be protected using one or more encryption techniques, such as those techniques provided by the Advanced Encryption Standard (AES), which superseded the Data Encryption Standard (DES), the IEEE 802.1 standard for port-based network security, pre-shared key, Extensible Authentication Protocol (“EAP”), Wired Equivalent Privacy (“WEP”), Temporal Key Integrity Protocol (“TKIP”), Wi-Fi Protected Access (“WPA”), and the like.

[0165] The Internet Protocol (“IP”) is the principal communications protocol in the Internet

protocol suite for relaying datagrams across network boundaries. Its routing function enables internetworking, and essentially establishes the Internet. IP has the task of delivering packets from the source host to the destination host solely based on the IP addresses in the packet headers. For this purpose, IP defines packet structures that encapsulate the data to be delivered. It also defines addressing methods that are used to label the datagram with source and destination information. [0166] The Transmission Control Protocol (“TCP”) is one of the main protocols of the Internet protocol suite. It originated in the initial network implementation in which it complemented the IP. Therefore, the entire suite is commonly referred to as TCP/IP. TCP provides reliable, ordered, and error-checked delivery of a stream of octets (bytes) between applications running on hosts communicating via an IP network. Major internet applications such as the World Wide Web, email, remote administration, and file transfer rely on TCP, which is part of the Transport Layer of the TCP/IP suite.

[0167] Transport Layer Security, and its predecessor Secure Sockets Layer (“SSL/TLS”), often runs on top of TCP. SSL/TLS are cryptographic protocols designed to provide communications security over a computer network. Several versions of the protocols find widespread use in applications such as web browsing, email, instant messaging, and voice over IP (“VoIP”). Websites can use TLS to secure all communications between their servers and web browsers.

[0168] The system **200** can further include a memory **214**. The controller **210** can interact with the memory **214** in order to perform operations such as determining and/or monitoring the location of the one or more telescoping riser assemblies **202** via the one or more electronic locators **209**. The memory **214** includes, in some embodiments, a program storage area and/or data storage area. The memory **214** can comprise read-only memory (“ROM”, an example of non-volatile memory, meaning it does not lose data when it is not connected to a power source) or random access memory (“RAM”, an example of volatile memory, meaning it will lose its data when not connected to a power source). Nonlimiting examples of volatile memory include static RAM (“SRAM”), dynamic RAM (“DRAM”), synchronous DRAM (“SDRAM”), etc. Examples of non-volatile memory include electrically erasable programmable read only memory (“EEPROM”), flash memory, hard disks, SD cards, etc. In some embodiments, the controller **110** or other processing unit, such as a processor, a microprocessor, or a microcontroller, is connected to the memory **214** and executes software instructions that are capable of being stored in a RAM of the memory **214** (e.g., during execution), a ROM of the memory **214** (e.g., on a generally permanent basis), or another non-transitory computer readable medium such as another memory or a disc. The memory **214** can be used to store information such as the location(s) of each of the one or more telescoping riser assemblies **202**.

[0169] According to some embodiments, the system **200** can include a database **216**. A database is a structured set of data typically held in a computer. The database **216**, as well as data and information contained therein, need not reside in a single physical or electronic location. For example, the database **216** may reside, at least in part, on a local storage device, in an external hard drive, on a database server connected to a network such as the network **213**, on a cloud-based storage system, in a distributed ledger (such as those commonly used with blockchain technology), or the like. The database **216** can be used to store information such as the location(s) of each of the one or more telescoping riser assemblies **202**.

[0170] It should be noted that each component described herein to be part of the system **200** can be operationally connected such that each component is able to communicate with the other components.

[0171] FIGS. **33** and **34** show perspective views of an agricultural field **300** that includes at least one telescoping riser assembly **302** and at least one raised area **304**. FIG. **33** shows a portion of the field **300** and FIG. **34** shows another portion of the field **300** wherein the raised area **304** is the same in both FIGS. **33** and **34**.

[0172] The agricultural field **300** of FIGS. **33** and **34** can be any suitable agricultural field. Each

telescoping riser assembly **302** shown in FIGS. **33** and **34** can be the telescoping riser assembly **10** described herein. Each telescoping riser assembly **302** shown in FIGS. **33** and **34** are different telescoping riser assemblies configured in the same agricultural field **300**.

[0173] The agricultural field **300** can comprise one or more telescoping riser assemblies **302**. For example, the agricultural field of FIGS. **33** and **34** could include six telescoping riser assemblies **302** according to various embodiments. The agricultural field **300** could comprise zero to N telescoping riser assemblies **302** wherein N is any number greater than zero.

[0174] The agricultural field **300** can comprise one or more raised areas **304**. The agricultural field **300** of FIGS. **33** and **34** shows a single raised area **304**. The raised area **304** could be any sort of projection of the ground such as a levy, berm, mound, heap, and the like. Each of the telescoping riser assemblies **302** can be configured in particular locations around the raised area **304**. Each telescoping riser assembly **302** can be any suitable distance from the raised area **304**. According to some embodiments, each telescoping riser assembly **302** can be positioned forty feet from the raised area **304**. According to some embodiments, the portion(s) of the agricultural field **300** that are not raised area(s) are lower than the raised area(s) **304** forming valleys and/or low areas. Each of the telescoping riser assemblies **302** can be positioned in such a valley and/or low area. Each valley and/or low area can include one or more telescoping riser assemblies **302**. For example, according to some embodiments, six telescoping riser assemblies **302** can be positioned in each valley and/or low area of the agricultural field **300**. According to some embodiments, the raised area(s) **304** can be constructed so that water can be directed to collect in the valley(s) and/or low area(s) of the agricultural field **300** wherein the water can be drained by the telescoping riser assemblies **302**. Each of the telescoping riser assemblies **302** in the agricultural field **300** can be spaced to synergistically work together to drain water and/or control erosion.

[0175] It should be noted that any slidable connection mentioned herein may be established through the use of one or more suitable slidable elements, including: a friction fit on a slip (non-stick) surface, guides, tracks, piston(s), shaft(s), sleeve(s), collar(s), ball bearing(s), actuator(s), linkage(s), pivot(s), and/or the like. According to some embodiments, any slidable connection mentioned herein only allows a component to slide in one dimension (i.e. linearly) with respect to another component. Yet, it is to be appreciated some minor relief in a second dimension can be to mitigate wear, tear, or failure of slidable components and/or elements. To that end, oils, grease, lubricants, antistatic agents, and/or other non-viscous fluids or devices can be applied where wear and tear is expected to further mitigate the same over time.

[0176] It should be mentioned that any connection, attachment, and/or fastening mentioned herein may make use of screw(s), nut(s), bolt(s), pin(s), rivet(s), staple(s), washer(s), grommet(s), latch(es) (including pawl(s)), ratchet(s), clamp(s), clasp(s), flange(s), tie(s), adhesive(s), weld(s), any other known fastening mechanism(s), or any combination thereof to facilitate fastening, connection, and/or attachment.

[0177] Therefore, as understood from the present disclosure, the drain riser assembly/telescoping riser assembly described herein provides proper water drainage from an agricultural field while simultaneously improving a farmer's effectiveness and efficiency in performing agricultural activities and also improving a farmer's financial bottom line as compared to the prior art. The telescoping riser assembly allows a farmer to not have to navigate and/or steer around a riser extending up from the ground when performing an agricultural activity. For example, when performing an agricultural activity, the farmer can retract the top riser of each telescoping riser assembly into the underground piping and/or into the ground such that the farmer does not need to navigate agricultural equipment around the riser assembly when performing an agricultural activity. This saves the farmer time, effort, and money. The farmer can perform the agricultural activity with fewer passes through the agricultural field while maintaining a wide swath which leads to decreasing the time and fuel needed to perform said agricultural activity. Therefore, the farmer can cut costs related to fuel consumption and/or wages paid to hired laborers. Additionally, the farmer

can maximize revenue by maximizing yield. If the farmer had to navigate around the riser assemblies, such as in the prior art, this would lead to a decrease in yield based on the fact that the agricultural activity (such as planting, harvesting, and the like) would be negatively affected due to having curved and/or missing rows. Thus, since the top riser of the telescoping riser assembly described herein can be retracted into underground piping and, thus, does not need to be navigated around when performing an agricultural activity, crops can be planted in straight rows such that the farmer's yield is not negatively affected, but rather, is maximized. Maximization of yields leads to maximization of revenue.

[0178] Additionally, the telescoping nature of the top riser of the riser assembly combined with the fact that the riser assembly includes a main cap, provides protection for aspect(s) of the riser assembly as well as prevents debris from entering aspect(s) of the riser assembly. When the riser assembly is retracted into the underground piping, the main cap is flush with the ground and/or fits firmly to the ground and acts as a sort of shield. Thus, aspects of the telescoping riser assembly, including the underground piping, are protected from being damaged by agricultural equipment when an agricultural activity is performed. Additionally, the telescoping nature of the top riser combined with the main cap of the riser assembly prevents debris from entering aspect(s) of the riser assembly including the underground piping. Unwanted debris entering the riser assembly can cause the riser assembly to operate less effectively and/or can cause damage to aspects of the riser assembly. Therefore, preventing debris from entering aspects of the riser assembly saves time, effort, and money in terms of cleaning debris out of aspects of the riser assembly, repairing aspects of the riser assembly, and/or replacing aspects of the riser assembly. Additionally, debris in the telescoping riser assembly and/or damage to aspects of the telescoping riser assembly can cause the telescoping riser assembly to not function properly and/or to not function to the best of its ability. This can lead to poor water drainage which can have negative effects on yields ultimately resulting in negative effects on the farmer's revenue. Thus, the telescoping riser assembly described herein provides advantages over the prior art.

[0179] Furthermore, the telescoping riser system described herein allows a farmer to determine and/or monitor the location of each telescoping riser assembly that is part of the system. Thus, a farmer can pinpoint the exact location of each telescoping riser assembly in a particular agricultural field and/or across multiple agricultural fields. The farmer can use any sort of HMI to determine where each telescoping riser assembly is located. Therefore, when the farmer wants to extend and/or retract the telescoping riser assemblies in a particular field, much time, effort, and money is saved by knowing the location of each telescoping riser assembly.

[0180] It is to be appreciated that the telescoping riser system described herein can be used with any number of orifices in either the bottom riser or the top riser, depending on the application. For example, the bottom riser has historically included an orifice to comply with governmental regulations. This configuration can thus still be used. However, it may be more beneficial to eliminate the orifice entirely so as to not restrict water flow in the area in which the orifice used to be. Moreover, the orifice can be beneficially employed in areas of the top riser where water is desired to be restricted, without needlessly restricting water flow in a way which would cause a detrimental effect such as a partial or total blockage of intended fluid flow.

[0181] Furthermore, the telescoping riser system described herein can include indicators that suggest to farmers how to use and/or install the telescoping riser system. For example, installation instructions can be included at the time the telescoping riser system is sold. Additionally or in lieu thereof, visual indicators such as arrows, text, and the like, can be included on one or more components (such as the top cap) to indicate a particular step for installation or use (such as attaching the bottom riser to the top riser, indicating a direction of locking or unlocking, etc.).

[0182] From the foregoing, it can be seen that the present disclosure accomplishes at least all of the stated objectives.

LIST OF REFERENCE CHARACTERS

[0183] The following table of reference characters and descriptors are not exhaustive, nor limiting, and include reasonable equivalents. If possible, elements identified by a reference character below and/or those elements which are near ubiquitous within the art can replace or supplement any element identified by another reference character.

TABLE-US-00001 TABLE 1 List of Reference Characters 10 telescoping riser assembly/drain riser assembly 12 cylindrical member/top riser 13 first end of cylindrical member 14 vibration isolator 15 second end of cylindrical member 16 snap cap 17 bushing 18 main cap 20 first cut line 22 second cut line 24 first section of cylindrical member 26 second section of cylindrical member 28 third section of cylindrical member 30 protrusion of snap cap 32 aperture of cylindrical member 34 overmold of snap cap 36 connection member of main cap 38 length of first section of cylindrical member 40 combined length of first and second sections of cylindrical member 42 combined length of first, second, and third sections of cylindrical member/total length of cylindrical member 44 guide slot of cylindrical member 46 locking mechanism 47 diameter of cylindrical member 48 back of guide slot 49 diameter of guide slot 50 body of bushing 51 first end of bushing 52 second end of bushing 53 tab(s) of bushing 54 flange(s) of bushing 55 inner surface of bushing 56 rib of bushing 57 outer surface of bushing 58 rim of bushing 60 diameter of bushing (inclusive of the rim) 62 length/height of bushing 64 side wall(s) of snap cap 66 top of snap cap 68 bottom of snap cap 70 slot(s) of snap cap 71 diameter of snap cap 72 height of side wall(s) 74 total height of snap cap 76 body of main cap 78 top of the body of the main cap 80 bottom of the body of the main cap 82 raised portion of connection member 84 plate of connection member 86 leg(s) of connection member 87 end member(s) of each leg of connection member 88 diameter of main cap 90 diameter of plate of connection member 92 height of main cap 94 underground piping/bottom riser 95 length of underground piping/bottom riser 96 top of underground piping/bottom riser 97 bottom of underground 98 diameter of underground piping/bottom riser 99 tab of underground piping/bottom riser 100 aperture of underground piping/bottom riser 102 large separator 104 medium separator 106 small separator 108 inner wall of underground piping/bottom riser 110 outer wall of underground piping/bottom riser 200 telescoping riser system 202 one or more telescoping riser assemblies 204 human-machine interface (HMI) 206 user interface (UI) 208 positioning system 209 one or more electronic locators 210 controller 212 component(s) for communication 213 network 214 memory 216 database 300 agricultural field 302 telescoping riser assembly 304 raised area

Claims

1. A drain riser assembly for use with water drainage from an agricultural field, the assembly comprising: a cylindrical member; a vibration isolator configured to slidably interact with the cylindrical member; a snap cap removably attachable to one end of the cylindrical member; and a main cap removably attachable to the snap cap.
2. The assembly of claim 1, wherein the cylindrical member comprises opposing guide slots.
3. The assembly of claim 2, wherein the vibration isolator is a removable bushing comprising opposing tabs.
4. The assembly of claim 3, wherein the opposing guide slots of the cylindrical member are configured to slidably engage the opposing tabs of the bushing.
5. The assembly of claim 3, wherein the bushing comprises a body having an outer surface and an inner surface, further wherein the bushing is configured to engage underground piping via the outer surface, and/or flange(s) thereof, and slidably interact with the cylindrical member via the inner surface.
6. The assembly of claim 5, wherein when the bushing is engaged with the underground piping, the cylindrical member is capable of sliding relative to the bushing such that the cylindrical member can telescopically extend from and retract into the underground piping, further wherein a body of

the main cap fits firmly to the ground when the main cap is attached to the snap cap, the snap cap is attached to the cylindrical member, and the cylindrical member is fully retracted into the underground piping.

7. The assembly of claim 2, wherein the snap cap comprises opposing slots wherein the slots of the snap cap are configured to slidingly interact with a back side of the guide slots of the cylindrical member.

8. The assembly of claim 7, wherein the snap cap further comprises opposing protrusions and the cylindrical member comprises two or more apertures, further wherein each of the opposing protrusions are configured to engage one of the two or more apertures when the snap cap is attached to the cylindrical member.

9. The assembly of claim 1, wherein the cylindrical member comprises a locking mechanism that allows the cylindrical member to be locked in place relative to the vibration isolator.

10. The assembly of claim 1, wherein the snap cap comprises an overmold and the main cap comprises a connection member, further wherein the overmold and the connection member can matingly engage to attach the snap cap to the main cap.

11. The assembly of claim 1, wherein the cylindrical member comprises first, second, and third sections, further wherein the first and second sections are separated via a first cut line and the second and third sections are separated via a second cut line.

12. The assembly of claim 1, wherein the vibration isolator comprises one or more tabs integrally formed with underground piping.

13. The assembly of claim 12, wherein the cylindrical member is capable of sliding relative to the one or more tabs integrally formed with the underground piping such that the cylindrical member can telescopically extend from and retract into the underground piping, further wherein a body of the main cap fits firmly to the ground when the main cap is attached to the snap cap, the snap cap is attached to the cylindrical member, and the cylindrical member is fully retracted into the underground piping.

14. A telescoping riser assembly for use with water drainage from an agricultural field, the assembly comprising: a bottom riser positioned generally underground; a vibration isolator configured to interact with or be integrally formed with the bottom riser; and a top riser configured to slidingly interact with the vibration isolator to extend from and retract into the bottom riser in a telescopic manner.

15. The assembly of claim 14, wherein the vibration isolator is a removable bushing configured to interact with the bottom riser.

16. The assembly of claim 14, wherein the vibration isolator comprises one or more tabs integrally formed with the bottom riser.

17. The assembly of claim 14, wherein the bottom riser and the top riser are generally cylindrical.

18. A telescoping riser system for use with water drainage from an agricultural field, the system comprising: at least one drain riser assembly of claim 1, wherein each of the at least one drain riser assemblies is positioned in an agricultural field.

19. The system of claim 18, further comprising a human-machine interface (HMI) operationally connected to each of the at least one drain riser assemblies.

20. The system of claim 19, further comprising a positioning system operationally connected to each of the at least one drain riser assemblies and/or to the HMI to determine and/or monitor the location of each of the at least one drain riser assemblies.
